

1 Introduction

1.1 Observations for masting-seed predator/flowering

- Define masting

1.2 Major hypotheses for masting

1.3 Hypotheses operate at different reproductive stages

- Predation satiation—seed dispersal
- Pollination coupling—pollination
- Resource matching—across stages
- Mechanisms are not mutually exclusive

1.4 Trait-based perspective

- Lit review on masting and traits
- Briefly connect traits to mechanisms
- Shift from seed and pollination to more resource matching oriented

1.5 Introduce framework for hypothesis x traits x masting

- Here we use North America tree data to test this framework
 - Masting info available
 - Have both masting and non-masting species

2 Discussion

2.1 Summary of main findings

- Seed weight supports predation satiation
- Pollination supports pollination coupling
- In the context with lit, discuss potential mechanism
- Transition to alternative hypotheses

2.2 Masting is a population level phenomenon

- Examples of species only mast in certain environment
- Traits need to be measured locally

2.3 Different species selected under different hypotheses

- Introduce flower masting, seed masting
- Different species, different mechanisms
- Finer definition of masting and more flowering traits needed
- Transition to phylogeny

2.4 Evolutionary scale

- Angiosperm and gymnosperm results
 - More masting gymnosperm sp
 - Leaf longevity longer for gymnosperm sp
 - Wind pollination for all gymnosperm sp
- Importance of testing within clades (*Quercus*)